

Zap/Fizzle

Class One

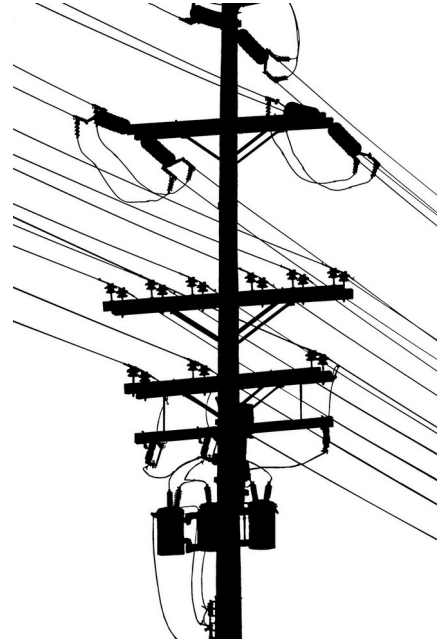
**★ Electronics Vs.
Electricity
Basic Electrical
Theory
Basic Components
Basic Circuits**

Electricity Vs. Electronics

Electrical Systems:

- Use electricity to directly run a machine
- Transport potential energy

Electricity Vs. Electronics



Electricity Vs. Electronics

Electronic Systems

- Use electricity to create a complex output based on input variables
- Interpretation of experience based on a framework

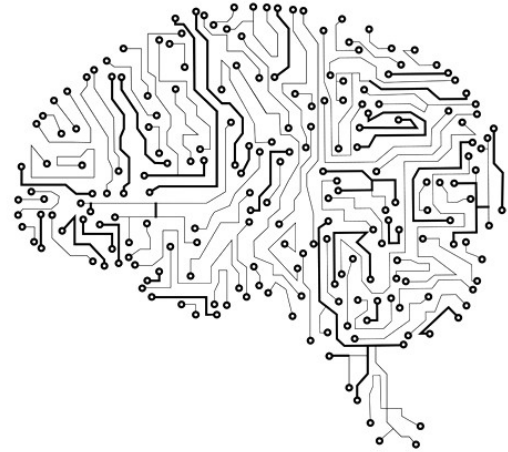
Electricity Vs. Electronics



Electricity Vs. Electronics



VS



Magic?

Carl Sagan Disagrees...

Electricity...

...is the presence and flow of molecular energy (static vs current)

...requires a clear pathway to flow

...gets converted to other types of *energy*

...is completely predictable

...is inextricably connected to magnetism

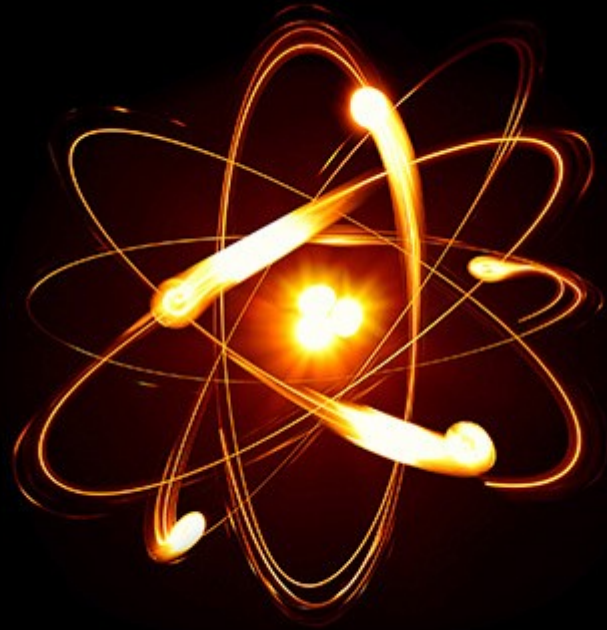


But Still...

HOW?

Atoms are the building blocks of the physical world.

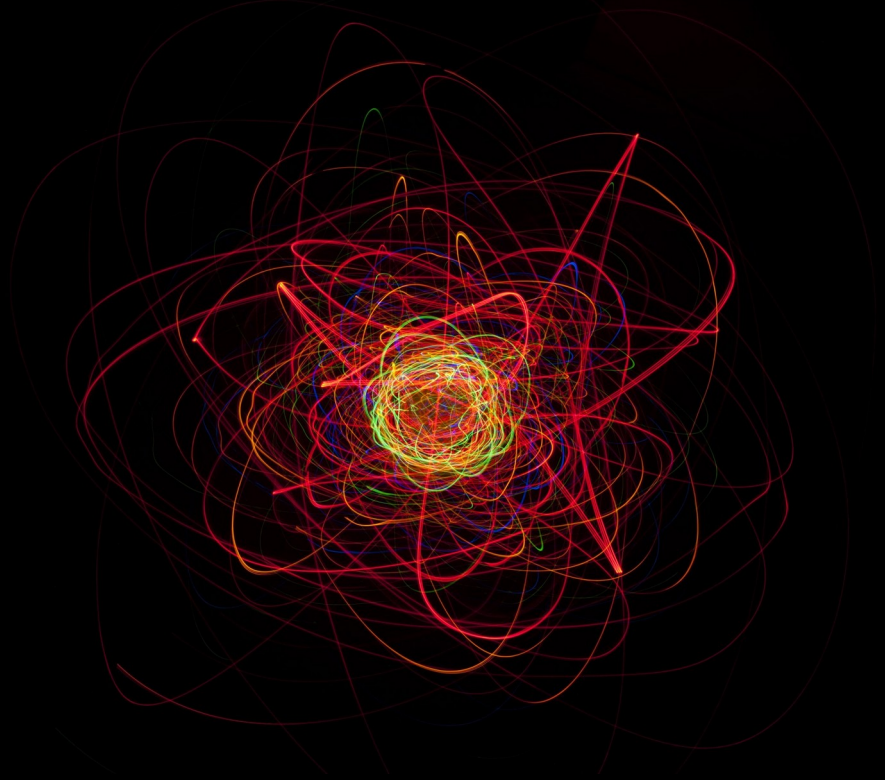
They are made of a centre, nucleus, and surrounded by a negatively charged cloud made of electrons.



But Still...

HOW?

In reality, they are not so neatly ordered.



But Still...

HOW?



I know exactly
where the electron
is, or how fast it's
going. Never both.

But Still...

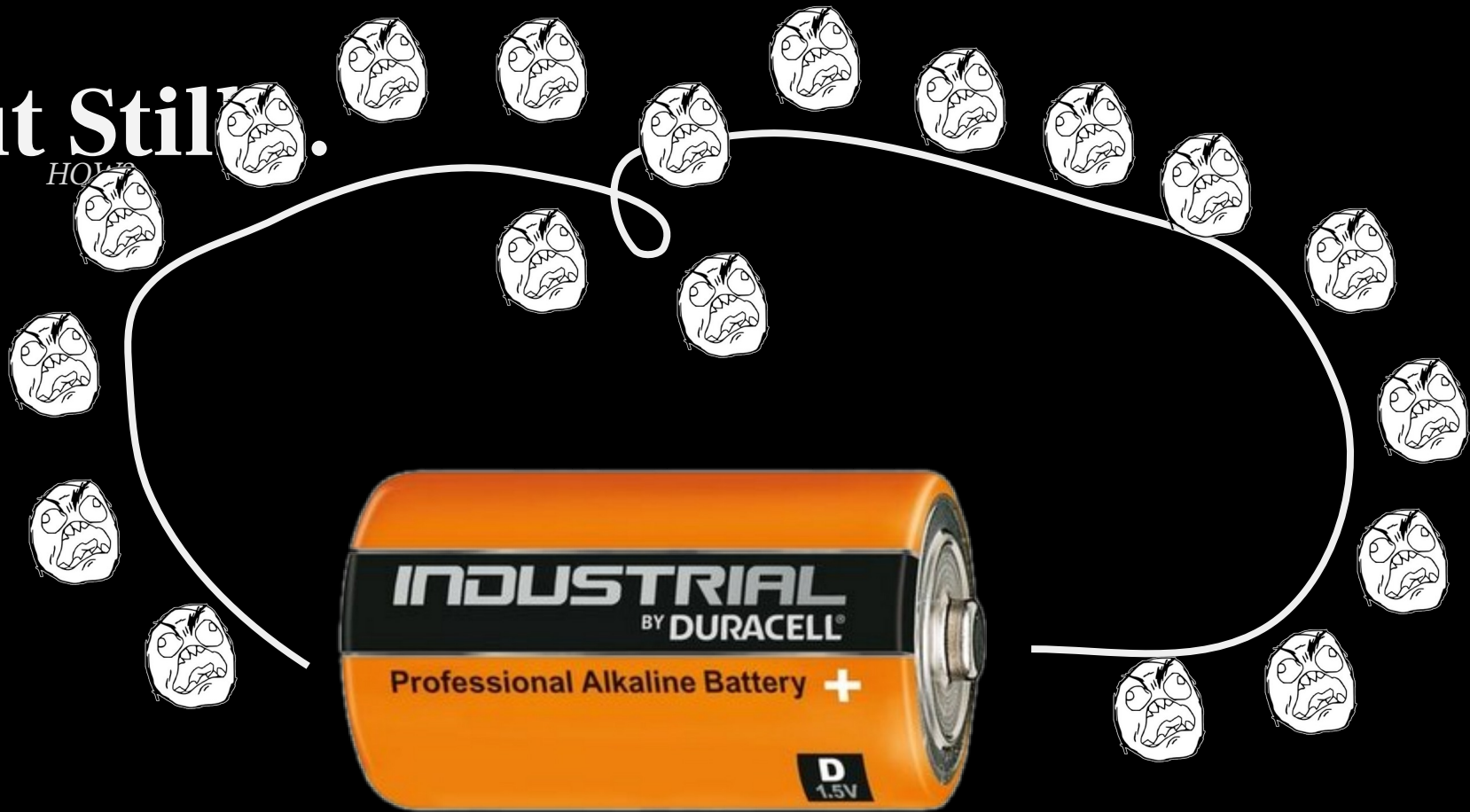
HOW?

Being so negative, electrons look for any excuse to get away.

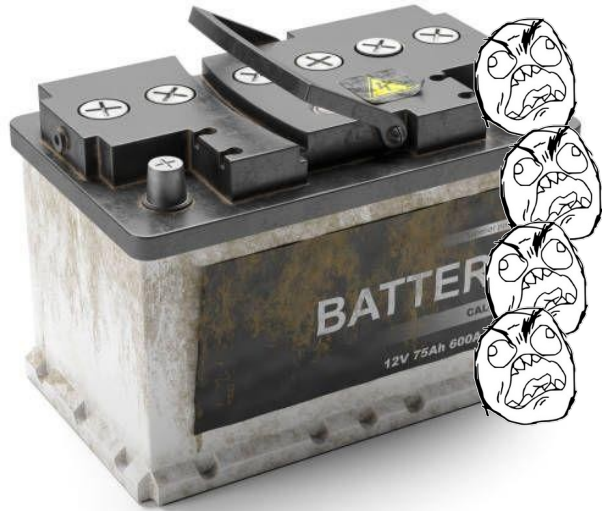
They move from atom to atom as space opens up for them.



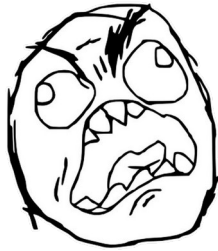
But Still.
HOW?



**Even though they are
Negative, Electrons still
follow rules**



W-whhaaa?



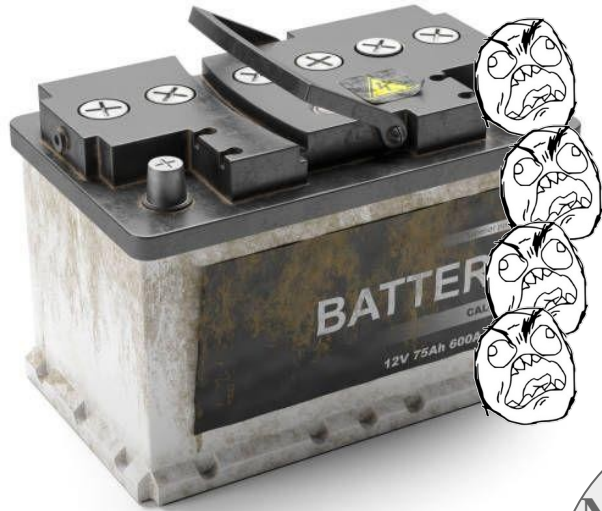
Fffff!
Fine.

**OBEY
PHYSICS!**



IT'S THE LAW!

Even though they are Negative, Electrons still follow rules



I gotta get away

Not without
us you ain't



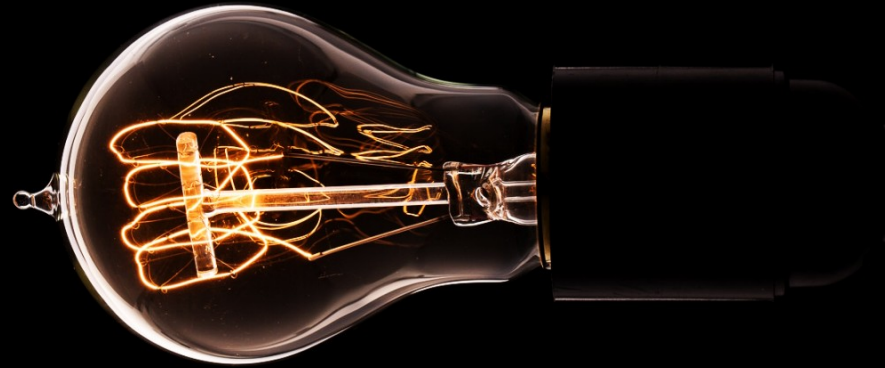
But Still...

HOW?

The movement of electrons flow through particular components, like a lightbulb.

This energy can then be converted into other things, heat, light, movement.

In electronics this movement causes complex machinery to operate, like molecular clockwork



In particul

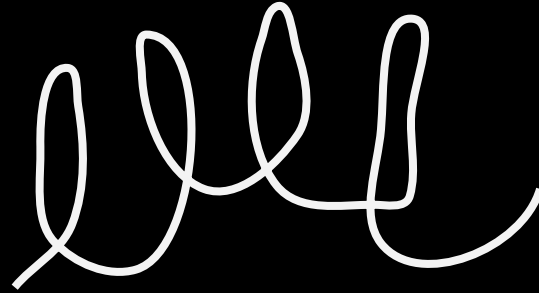
You still haven't told me how it works



In particular

You still haven't told me how it works

A Light bulb has a small wire running through it - the filament

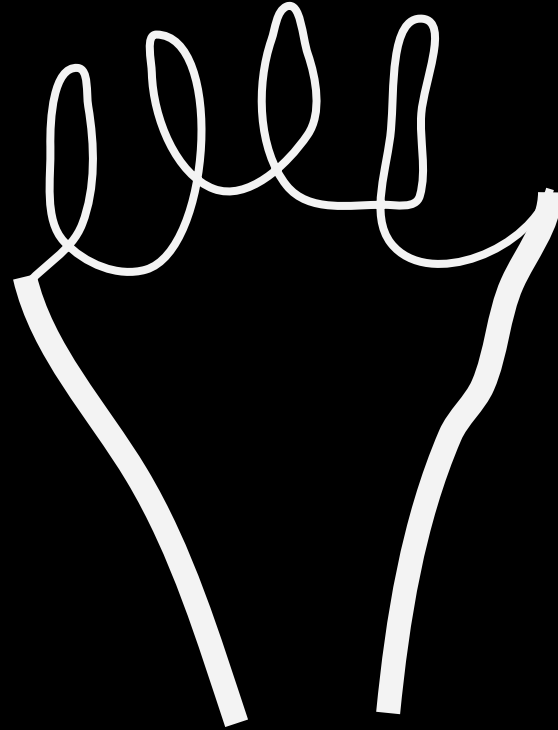


In particular

You still haven't told me how it works

A Light bulb has a small wire running through it - the filament

The leads coming into the filament are much thicker



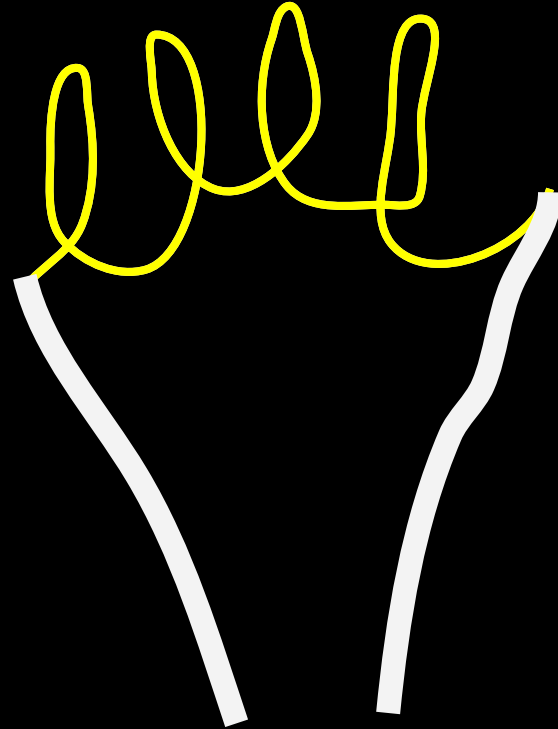
In particular

You still haven't told me how it works

A Light bulb has a small wire running through it - the filament

The leads coming into the filament are much thicker

The thinner filament causes friction with the electrons



In particul

You still haven't told me how it works



In particul

You still haven't told me how it works

The intense resistance on the electrons causes *heat*. This heat gives off *radiant energy*, running from visible spectrum light into IR. We can feel IR as heat on our skin. It's all part of the same spectrum of *electromagnetic energy*!



It doesn't burst into flames because there is no oxygen in the bulb - only inert gases (argon and nitrogen, usually)

In particular

You still haven't told me how it works



Electricity Vs. Electricity

AC=

Alternating
Current

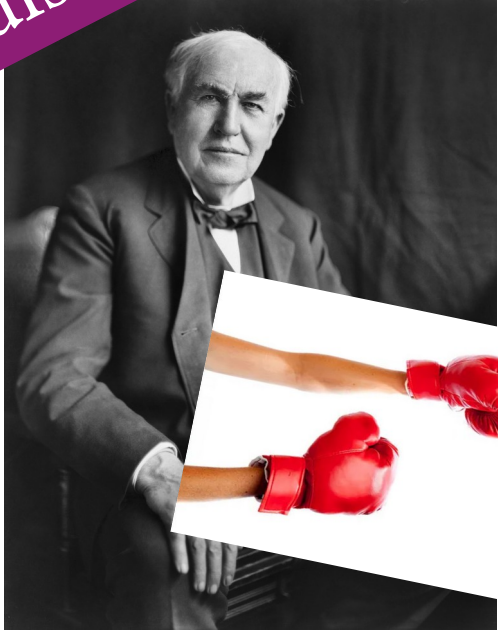


DC=

Direct Current

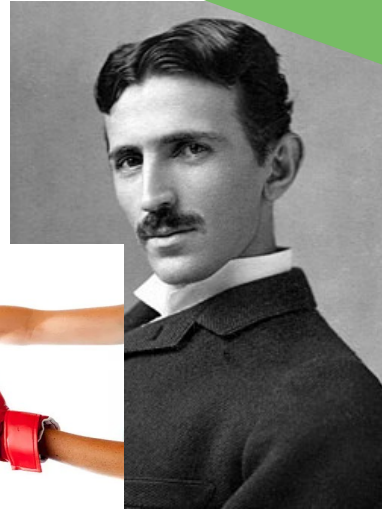
Electricity Vs. Electricity

Edison

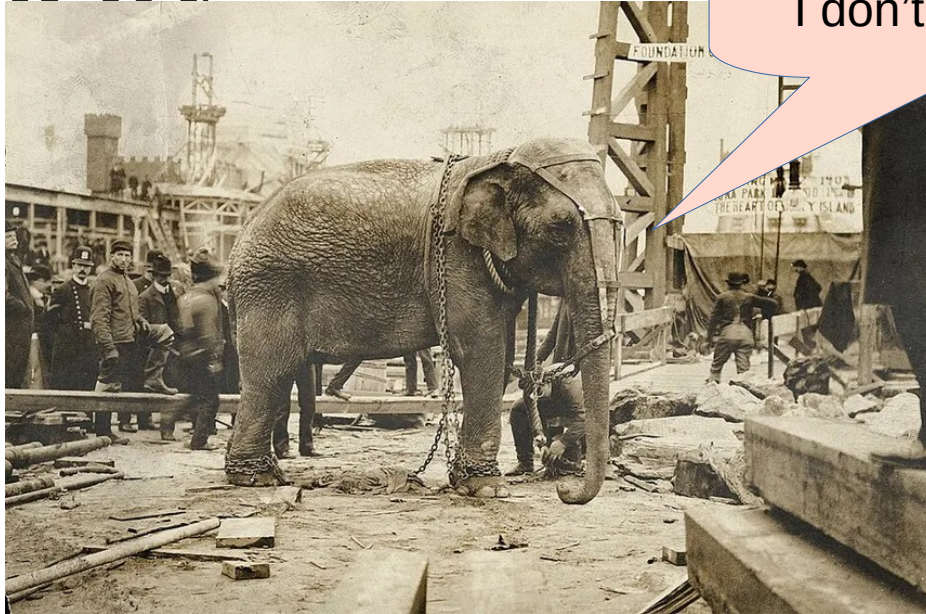


~Vs~

TESLA



Electricity Vs. Electricity



My Fate is sealed.
I don't blame Edison

Electricity Vs.

Why was Edison stupid?

Electricity



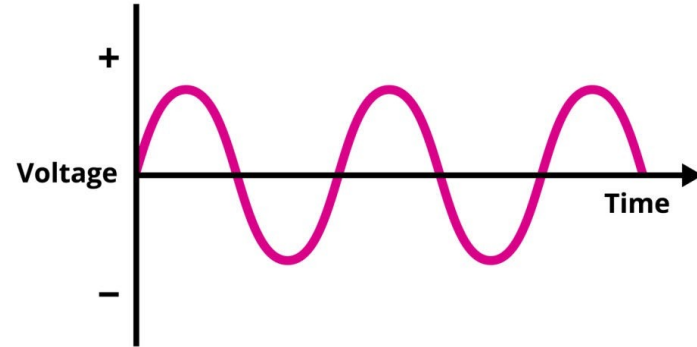
These would melt, carrying the current DC power requires to run a city. (Also he was just kind of a dick.)

Electricity Vs. *But what is it?* Electricity

Direct Current (DC)



Alternating Current (AC)



Electricity ♡

How do we make it work
for us?



I'm a flippin'
superhero!

A network diagram is constructed on a white surface using numerous silver pushpins as nodes. The nodes are interconnected by thin, translucent rubber bands in various colors including red, orange, yellow, green, and purple. The connections form a complex web of triangles and other polygons. A semi-transparent dark grey rectangle is overlaid on the left side of the image, containing the title and subtitle text.

Pathways

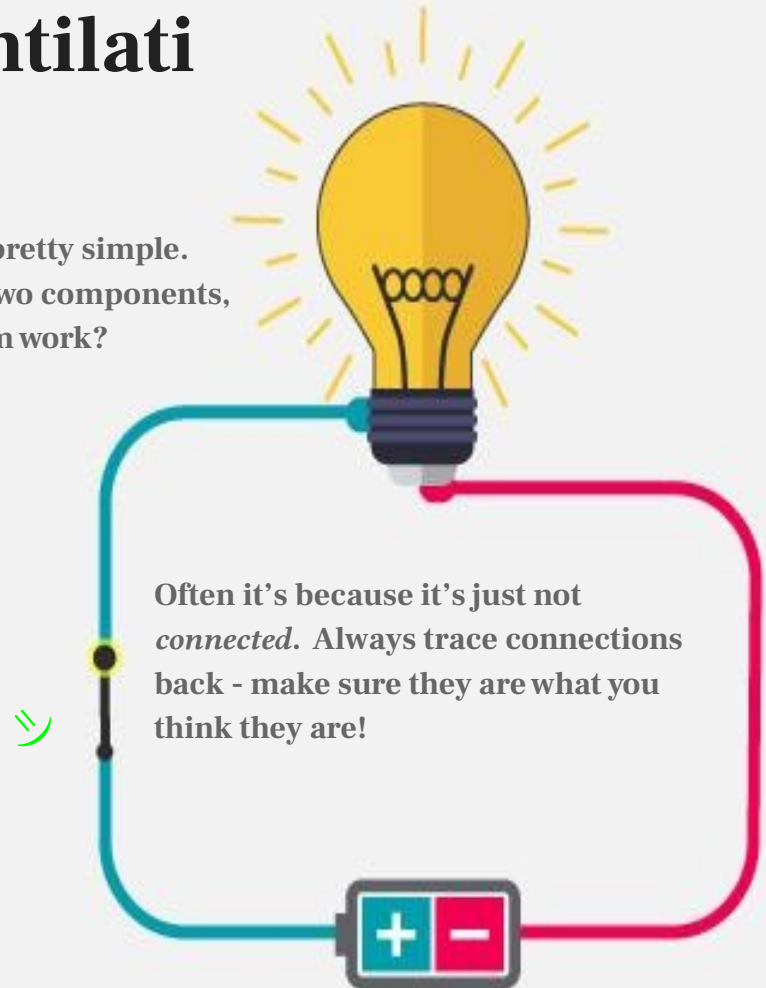
This is going to come up again and again.



PATHWAYS CAN BE VERY COMMPLEX

Hyperventilati ng

We're going to start pretty simple.
Think about one or two components,
how do we make them work?

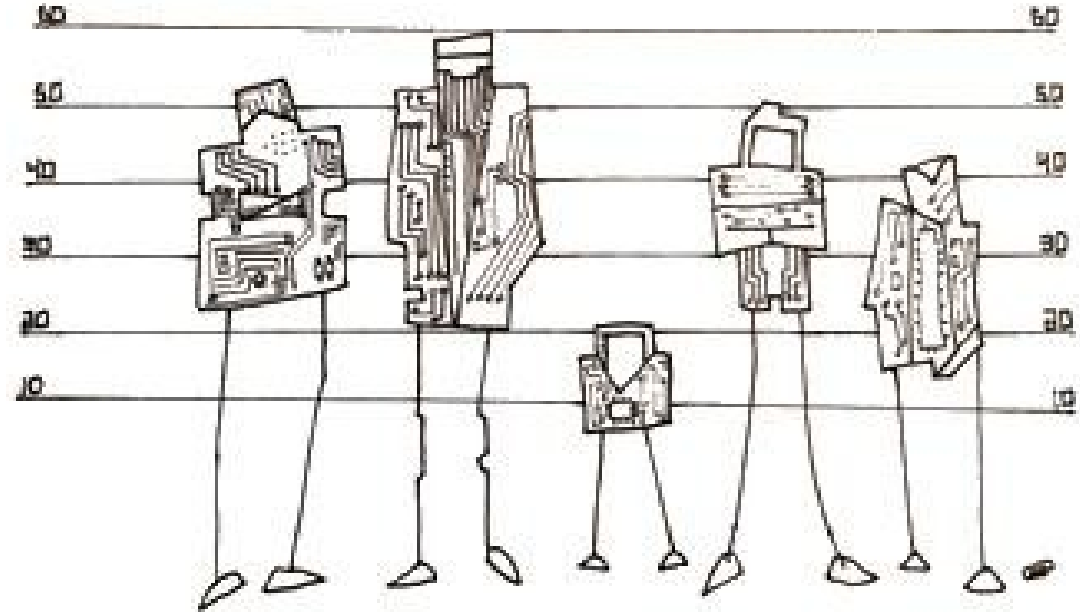


IT's the centre one

It's also possible the thing is connected, but electricity is finding a route around it - this is called a *short circuit*.

Like most of us, electricity will always take the easiest route. The path of least *resistance*.

But how does electricity move in a circuit?



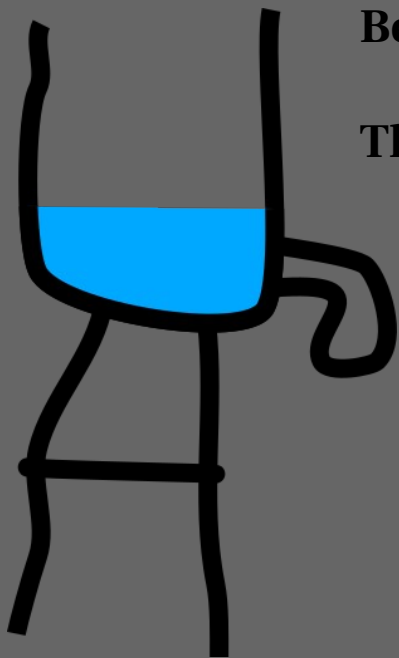
Which way would YOU rather go?

Electricity will ALWAYS find the shortest route to complete the circuit, even (or maybe especially) if it means bypassing the loads in a circuit.



The Water Analogy

There really is no escaping it



Behold! The world's crappiest watter tower!

The tap is off. No watter flows. This sucks.

The Water Analogy

There really is no escaping it



Behold! The world's crappiest watter tower!

The tap is off. No watter flows. This sucks.

Huzzah! The tap is on! Watter flows freely!

... Somewatt freely. There isn't much pressure. Flow is slow. Watt can we do?

The Water Analogy

There really is no escaping it



Double huzzah! Extra watter has increased pressure! Now there is a righteous splooge of watts. Watt? Watt is going on here?

The Water Analogy

There really is no escaping it



Currently we've increased the pipe. Moar watter!

I am so amped up right now! Splain how this all works!

The Water

The shocking results

Analogy



Volume/height of watter, affects pressure of splooge.

Force = **Volts**

Size of outlet, affects capacity of splooged at any given moment.

Amount able to pass through = **Amperes**

Resultant splooge.

Watts

The Water **Watts = Volts x Amps**

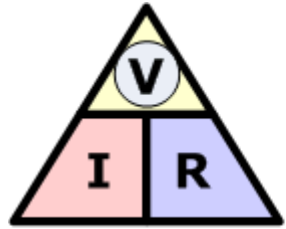
Ohm my god!

Analogy **Volts = Watts / Amps**

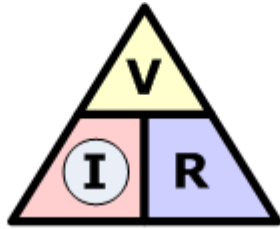
Amps = Watts / Volts

Simple math is simple!

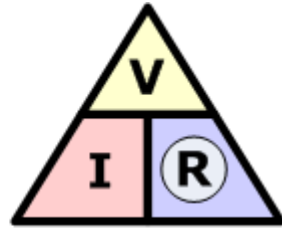
Mathematical!



$$\textcircled{V} = I \times R$$

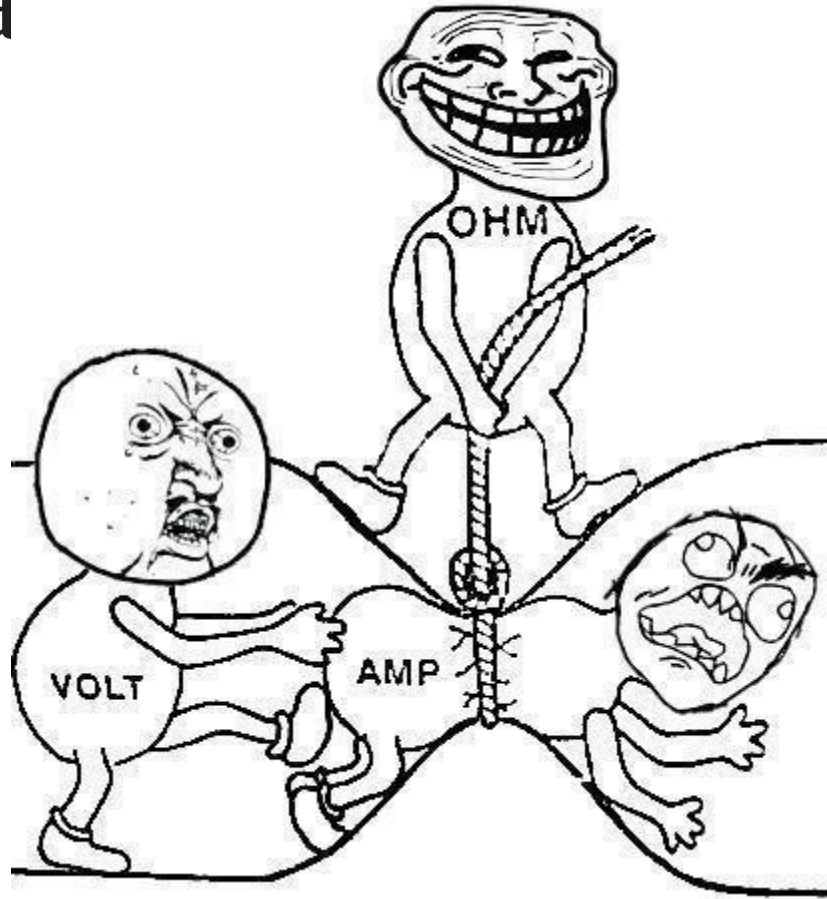


$$\textcircled{I} = \frac{V}{R}$$



$$\textcircled{R} = \frac{V}{I}$$

Mathematical



Bits ♦ Pieces

Stuff makes it Work

★ We know current flow requires a pathway.



Bits ♦ Pieces

Stuff makes it Work

- ★ We know current flow requires a pathway.
- ★ This pathway is usually made of highly conductive material.



Bits ♦ Pieces

Stuff makes it Work

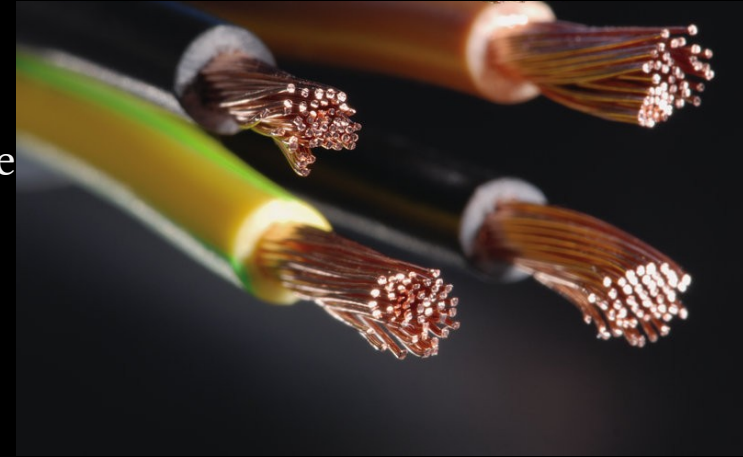
- ★ We know current flow requires a pathway.
- ★ This pathway is usually made of highly conductive material.
- ★ The opposite of conducting is insulating, this impedes the flow of current.



Bits ♦ Pieces

Stuff makes it Work

- ★ We know current flow requires a pathway.
- ★ This pathway is usually made of highly conductive material.
- ★ The opposite of conducting is insulating, this impedes the flow of current.
- ★ A wire is usually a highly conductive material encased in an insulator



Other

“Keepy safies” and “Get in the

Insulators

workies

* Vinyls, plastics, rubbers



Other

*“Keepy safies” and “Get in the
wiggles”*
★ Vinyls, plastics, rubbers

★ Fibrous material



Other

“Keepy safies” and “Get in the

Insulators

ways”
★ Vinyls, plastics, rubbers

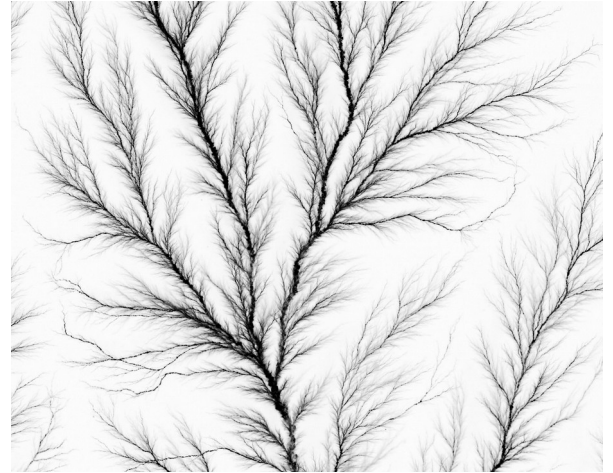
★ Fibrous material

★ Vitreous & ceramic material

Other

“Keepy safies” and “Get in the wires”
Insulators

- ★ Vinyls, plastics, rubbers
- ★ Fibrous material
- ★ Vitreous & ceramic material



Other

“Keepy safies” and “Get in the woggles”
★ Vinyls, plastics, rubbers

- ★ Fibrous material
- ★ Vitreous & ceramic material
- ★ Skin?



Electrical Pathways

Heart attack in a can

- ★ Basically any metal
- ★ ... Just add water
- ★ Can disrupt biological processes

Bits N Pieces

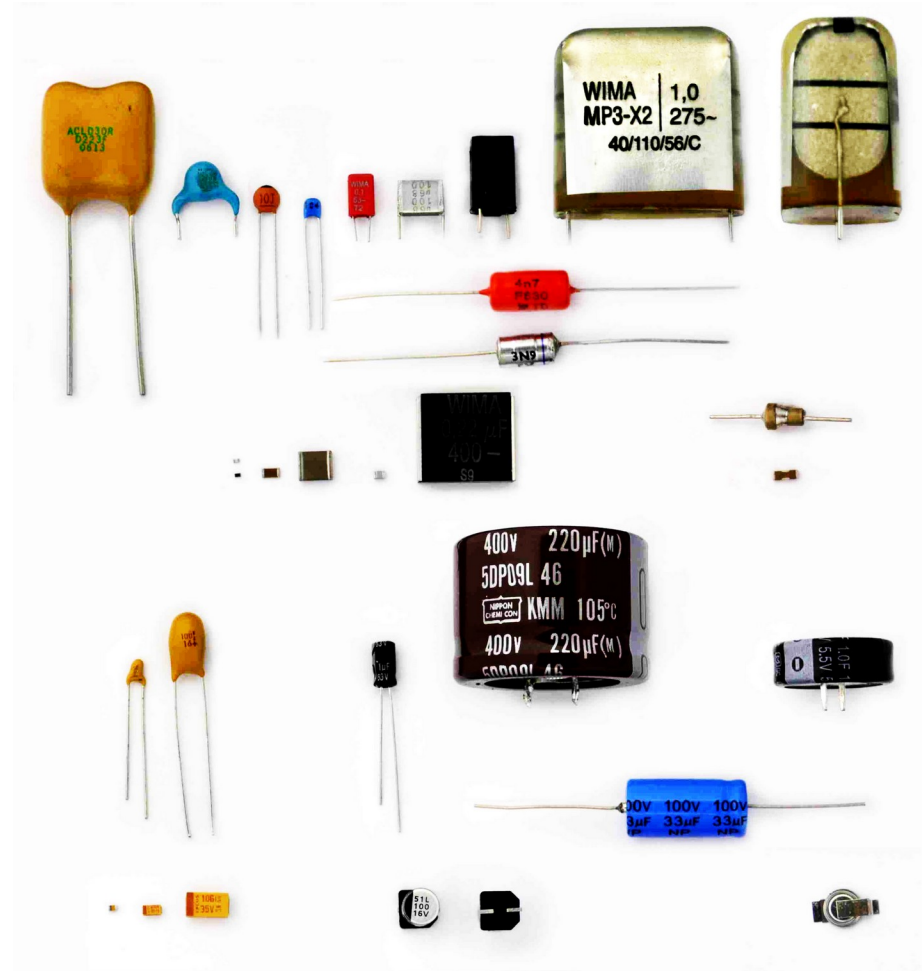
Like Lego, but more electrocutiony

- ★ **Resistors** reduce or redirect the current flow precisely.
- ★ **Colour** bands indicate value of resistance
- ★ Not polarized



Like Lego, but more electrocutiomy

★ Some, but not all polarized



Bits N Pieces

Like Lego, but more electrocutiony

- ★ **Switches** open or close electrical pathways (change its state)
- ★ Some have multiple states



Bits N Pieces

Like Lego, but more electrocutiony

- ★ **Batteries** store and provide running electricity
- ★ Always polarized



Bits N Pieces

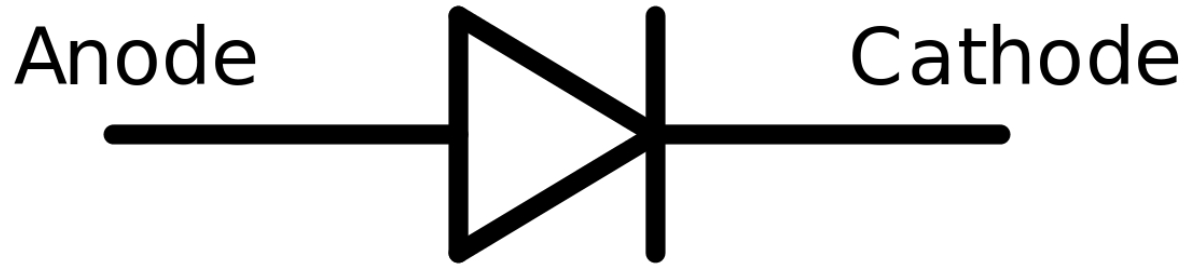
Like Lego, but more electrocutiony

- ★ **Diodes** allow current to pass in one direction only
- ★ Cover a huge range of functions including signal transmission, voltage protection and regulation...



DIODES

!!

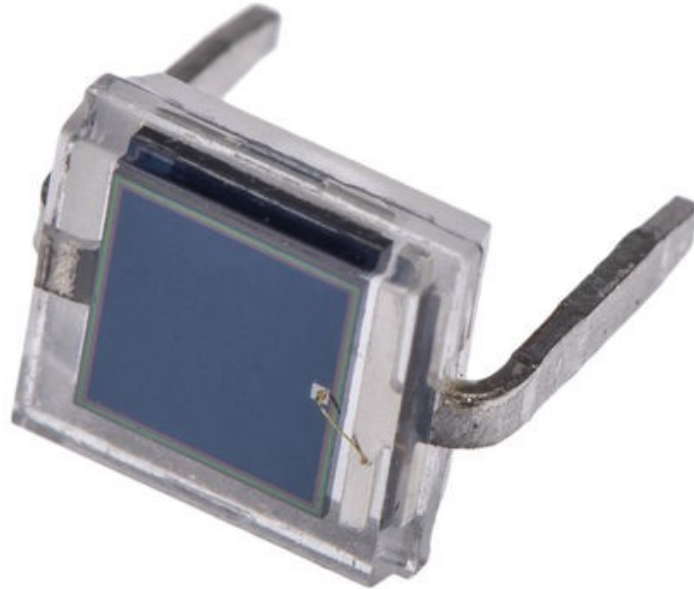


Allow electricity to flow in one direction only

DIODES



...some can also produce electrical current from light!



DIODES Light Emitting DIODES!!

...some can also produce light from electricity!

Yay

LEDs

